



Variations and asymmetries in regional brain surface in the genus *Homo*

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ABSTRACT

Paleoneurology is an important field of research within human evolution studies. Variations in size and shape of an endocrast help to differentiate among fossil hominin species whereas endocranial asymmetries are related to behavior and cognitive function. Here we analyse variations of the surface of the frontal, parieto-temporal and occipital lobes among different species of *Homo*, including 39 fossil hominins, ten fossil anatomically modern *Homo sapiens* and 100 endocrasts of extant modern humans. We also test for the possible asymmetries of these features in a large sample of modern humans and observe individual particularities in the fossil specimens.

This study contributes important new information about the brain evolution in the genus *Homo*. Our results show that the general pattern of surface asymmetry for the different regional brain surfaces in fossil species of *Homo* does not seem to be different from the pattern described in a large sample of anatomically modern *H. sapiens*, i.e., the right hemisphere has a larger surface than the left, as do the right frontal, the right parieto-temporal and the left occipital lobes compared with the contra-lateral side. It also appears that Asian *Homo erectus* specimens are discriminated from all other samples of *Homo*, including African and Georgian specimens that are also sometimes included in that taxon. The Asian fossils show a significantly smaller relative size of the parietal and temporal lobes. Neandertals and anatomically modern *H. sapiens*, who share the largest endocranial volume of all hominins, show differences when considering the relative contribution of the frontal, parieto-temporal and occipital lobes. These results illustrate an original variation in the pattern of brain organization in hominins independent of variations in total size. The globularization of the brain and the enlargement of the parietal lobes could be considered derived features observed uniquely in anatomically modern *H. sapiens*.

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R É S U M É

La paléoneurologie est un champ de recherche important dans le cadre des études sur l'évolution humaine. Les variations de taille et de forme de l'endocrâne sont en effet utiles pour différencier les différentes espèces d'homininés, alors que les asymétries cérébrales sont reliées au comportement et aux capacités cognitives. Nous analysons ici pour la première fois les variations de la surface des lobes frontaux, pariéto-temporaux et occipitaux chez plusieurs espèces du genre *Homo*, en se basant sur un matériel comprenant 39 homininés fossiles, dix Hommes anatomiquement modernes fossiles et 100 endocrânes d'Hommes actuels. Nous recherchons aussi les éventuelles asymétries pour ces caractères sur le large échantillon d'Hommes anatomiquement modernes et observons les particularités individuelles des spécimens fossiles. Cette étude apporte ainsi de nouvelles informations au sujet de l'évolution du cerveau au sein du genre *Homo*.

D'un point de vue méthodologique, nous démontrons que les recherches menées sur des endocrânes doivent être construites sur des références anatomiques directement reliées au cerveau, plutôt que sur des points situés sur la surface interne du crâne, si le propos est justement d'étudier la morphologie du cerveau. Par ailleurs, des divergences dans la variation des données de surface et de volume pour les Hommes actuels, observées entre cette étude et d'autres en imagerie médicale sur le vivant, illustrent

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que nous ne devrions jamais utiliser des données linéaires ou de surface obtenues sur des endocrânes, le seul matériel disponible pour les spécimens fossiles, pour interférer les éventuelles variations de volume des différentes zones cérébrales.

Par ailleurs, nos résultats illustrent que le schéma d'ensemble des asymétries pour les différentes zones cérébrales chez les espèces fossiles du genre *Homo* ne semble pas être différent de celui décrit pour l'échantillon large d'Hommes anatomiquement modernes, i.e., l'hémisphère droit a en moyenne une surface plus importante que le gauche, les lobes pariéto-temporaux droit et occipital gauche sont aussi plus développés que les zones correspondantes sur l'autre hémisphère.

Il apparaît aussi que les spécimens asiatiques attribués à l'espèce *Homo erectus* se différencient des autres échantillons du genre *Homo*, dont les spécimens africains et géorgiens qui sont aussi parfois inclus dans le taxon *Homo erectus*. En effet, les fossiles asiatiques ont une surface relative des lobes pariéto-temporaux significativement plus petite que les autres échantillons analysés dans le cadre de cette étude. Enfin, les Néandertaliens et les Hommes anatomiquement modernes, qui partagent les valeurs les plus importantes parmi tous les hominins pour le volume endocrânien, ont une structure du cerveau différente lorsque sont considérées les contributions relatives des lobes frontaux, pariéto-temporaux et occipitaux. Ces résultats illustrent des variations inédites dans le schéma d'organisation cérébrale chez les hominins, indépendamment de la taille d'ensemble du cerveau. La globularisation du cerveau et le développement des lobes pariéto-temporaux semblent pouvoir être considérés comme des caractères dérivés, propres aux Hommes anatomiquement modernes.

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Introduction

Paleoneurology is an important field in the study of human evolution as it contributes important information to other areas of paleoanthropology. Variations in the size and shape of an endocranial cast help in differentiating among fossil hominin species. Moreover, anatomical traits and asymmetries of the endocranial cast are also related to behavior and cognitive function.

Global shape of the endocranial cast has been frequently analysed through linear metrics or geometric morphometrics (e.g., Holloway, 1981a; Grimaud-Hervé, 1997; Falk et al., 2000; Bruner et al., 2003; Balzeau, 2005, 2007; Bruner and Holloway, 2010; Wu et al., 2010, 2011; Bienvenu et al., 2011). A trend toward brain size increase during hominin evolution is well documented and established (e.g., Leakey et al., 1964; Holloway, 1972; Rightmire, 2004; Tobias, 2006; Sousa and Wood, 2007), with some exceptions such as *Homo floresiensis* (e.g., Falk et al., 2005, 2009; Aiello, 2010) or global size reduction in *Homo sapiens* (Henneberg, 1998; Grimaud-Hervé et al., 2011). Some reorganizational changes are also observed (e.g., Dart, 1956; Holloway et al., 2004), involving sulcal patterning, gyral extension and cytoarchitectural areas (e.g., Holloway, 2008), as well as variations in meningeal patterns (e.g., Weidenreich, 1938; Falk and Conroy, 1983; Falk, 1983, 1993; Saban, 1984, 1995; Grimaud-Hervé, 1997). However, information about variation in global size of the different lobes and their relationship to each other is particularly scarce for hominins. This, in particular, is due to the difficulty in precisely delimiting the various lobes on endocranial casts. Moreover, the quantification of their surfaces is difficult. Previous works have proposed quantification of the volume or regional surface areas of the endocranial casts in fossil hominins (e.g., Weaver, 2005; Wu and Pan, 2011). However, such an approach is obviously limited by the impossibility of defining intracortical boundaries on endocranial material.

The emergence and occurrence of brain anatomical asymmetries have been extensively explored in the human fossil record to document the evolution of brain capacities and behavior (e.g., LeMay, 1976, 1999; Holloway and De La Coste-Lareymondie, 1982; Tobias, 1987; Balzeau et al., 2012). The study of brain surface asymmetries is difficult because it requires the definition of surface structural parameters and their homologs. The term 'petalia,' for instance, originally referred to the protrusions of one hemisphere beyond the other and is now widely used in reference to both protrusions and lobar asymmetries, or to cerebral asymmetries revealed by voxel-based morphometry (LeMay, 1976;

Holloway, 1981b, c; Holloway and De La Coste-Lareymondie, 1982; Hopkins and Marino, 2000; Good et al., 2001; Pilcher et al., 2001; Watkins et al., 2001; Holloway et al., 2004; Hopkins et al., 2008). Most of the studies of cerebral asymmetries in fossil hominins have addressed the protruding level or width of cerebral areas from a qualitative viewpoint.

The difficulties of paleoneurology reside in the complexity of the material being analysed (Grimaud-Hervé, 1997; Holloway et al., 2004). Details about sulci and gyri are not always well identifiable. Moreover, sulci do not always correlate precisely with the borders of functional or cytoarchitectonic areas. Landmarks on the internal surface of the skull, such as endobregma, endolambda or various foramina, are easier to identify and are sometimes preferred to landmarks pertaining to brain anatomy. Robust and reproducible methodologies dealing with direct indicators of brain anatomy need to be developed.

The number of brain anatomical traits, including structural asymmetries, observable on endocranial casts is limited. Nonetheless, this is the only available material for fossil specimens. Another limitation is the complex relationship between brain shape and endocranial shape (Fournier et al., 2011). Although the effects of age and sex are not uniform across brain regions, numerous cross-sectional as well as longitudinal neuroimaging studies reveal an overall significant age-related reduction of total brain volume (e.g., Resnick et al., 2000, 2003; Bartzokis et al., 2001; Scallan et al., 2003; Kochunov et al., 2005; Kruggel, 2006; Rettmann et al., 2006). Therefore, if the relationship between brain shape and brain age-related atrophy is difficult to appreciate, particularly in fossil species (Tobias, 2006), the endocranial cast shape in general reflects the shape of the brain of the young adult at the time of normal brain growth completion. Moreover, this relationship occurs in the context of the association of the shape of the brain and of the shape of the skull during the development (Moss and Young, 1960). Normal brain shape and brain shape asymmetries are therefore safely studied on endocranial casts. In addition, comparison is made between the same developmental stages corresponding to normal brain growth completion. This aspect is definitively an advantage when comparing specimens of different ages from different hominin species with most probably different modalities and rates of growth, development, and aging.

In this context, this study has two main analytical objectives. The first is to document variations in the surface of the frontal, parieto-temporal and occipital lobes among different species of the genus *Homo*. The second objective is to test for the possible

asymmetries of these features in a large sample of anatomically modern *H. sapiens* (AMH) and to observe individual particularities in the fossil specimens. **Therefore, we first analyse the possible correlation between endocranial volume and the surface of the frontal, parieto-temporal and occipital lobes in the genus *Homo*.** We also test possible variations among species and analyse the related evolution of the global shape of the brain. Finally, we discuss the evolution of the brain asymmetries and its correlation with brain capacities and behavior.

Materials

The analysed samples include physical endocasts of various hominin species (Table 1). The sample is comprised of 39 fossil hominins, ten fossil AMH and 100 endocasts of extant modern humans. The fossil specimens are all adult, except possibly KNM-WT 15,000, *Sinanthropus* III, Sambungmacan 3 and Gibraltar 2, Le Moustier, and Teshik Tash, which nevertheless certainly had achieved most of their normal brain growth (e.g., Guihard-Costa and Ramirez-Rossi, 2004; Balzeau, 2007).

The individuals in the extant sample were non-pathological adults. They all come from the anthropological collections of the Musée de l'Homme, Paris. Sex is not known for the complete sample and the geographic origin of these specimens is diverse. Most specimens are from recent historical times, with a large part from the last two centuries. The physical endocasts of the fossil hominins were analysed at Colombia University, USA, as well as at the Institut de Paléontologie Humaine and at the Musée de l'Homme, Paris, France.

Methods

The endocasts of the fossil hominins analysed here do not suffer from marked distortion and do not show large missing areas (Grimaud-Hervé, 1997; Holloway et al., 2004). Precise quantification of the surface of the various lobes was therefore possible. The measurement of surface area was conducted on physical endocasts by the same person (DGH) to avoid inter-observer errors. In order to

Table 1
Inventory of the analysed specimens with general information on the taxonomic attribution, geographical area and chronological age.

Species	Geographical area	Specimens	Age (Ma)
<i>Homo habilis</i> s.l.	Africa	KNM ER 1470, 1813, OH 13, 24	1.9–1.5
<i>Homo erectus</i> s.l.	Georgia	D2280	1.8
	Africa	KNM-ER 3733, 3883, KNM-WT 15000, OH 9	1.8–1.2
	China	Ckn. D1.PA.17 (II), E1.PA.16 (III), L1.PA.98 (X), L2.PA.99 (XI), L3.PA.100 (XII)	0.4
	Indonesia	Trinil 2, Sangiran 2, 10, 12, 17 Ngandong 1, 5, 6, 7, 11, 12, Sambungmacan 3	Around 1 0.2–0.05
Neandertals s.l.	Eurasia	Arago, Ehringsdorf H9, Reilingen, Swanscombe	0.4–0.2
		Neandertal, Gibraltar 2, La Chapelle aux Saints, La Ferrassie 1, La Quina H5, Le Moustier, Spy 1, 10, Teshik-Tash	0.1–0.03
Anatomically modern	Levant	Qafzeh 6, Skhul V	0.08
<i>Homo sapiens</i>	Europe	Brno III, Cro-Magnon 1, 3, Dolní Věstonice 2, Predmost 3, 4, 9, 10	0.03
	Extant sample	100 endocasts	

include as many specimens as possible, only the lateral and upper portions of the endocast were considered. These include the portions above the lateral border of the orbito-frontal cortex, the lateral inferior border of the temporal lobes and above the transverse sinuses. Thin and transparent paper was applied on the endocast, and kept in position by adhesive tape. The limits of the frontal, parieto-temporal and occipital lobes (Grimaud-Hervé, 1997) were drawn on this paper. The limit between the frontal (located anteriorly) and the parieto-temporal (located posteriorly) lobes correspond superiorly and medially to the junction of the central sulci and infero-laterally to the course of the small depressions toward the upper part of the sylvian valley. This limit runs posteriorly to the coronal suture following a different orientation, getting closer to the suture while approaching the sylvian valley. The limit between the parieto-temporal lobes (located anteriorly) and the occipital lobes (located posteriorly) correspond medially to the parieto-occipital sulcus, which is frequently identifiable anteriorly to the endolambda, and laterally to the pre-occipital notch. Globally, this limit follows the orientation of the lambdoid suture but does not correspond exactly to its course. The limits of the lobes were then defined by drawing continuous sulci around closed areas. These limits between the different lobes were distinguishable with enough confidence to be recognized independently by two of the authors. We did not separate the temporal and parietal lobes in our analyses because their limit above the second and third temporal convolutions is not delimited by the lateral sulcus that stops its course more anteriorly. The separation between these two lobes is therefore not identifiable in this area. We choose not to propagate posteriorly the course of this sulcus toward the parieto-occipital limit to avoid interpretations in the limits of the parietal and temporal lobes. The inter-hemispheric sulcus, the centre of the sagittal sinus and the mid-sagittal plane were used to separate the right and left lobes. The paper was cut along the line drawn to delineate the surface of each lobe, and pressed flat without any crease by notches cut around the edges. The paper tucked into the creases formed during the application of the paper to the endocasts was not included in the quantification of the endocranial surface. The surface was accurately measured in mm² with a digital planimeter (Haff 300–301). We chose to analyse physical endocasts instead of virtual replicas or virtually reconstructed endocasts for various reasons. Data have been compiled over years in different institutions, so the methodology was consistent across the complete samples. Moreover, small details on the endocranial surface corresponding to the sulci are easier to identify and to demarcate on physical models.

The validity of the employed protocol was tested by comparing the total surface measured with this method with the surface measured on virtual endocasts of the same individuals obtained via laser scanning or CT data. The quantification on the virtual models was made on five endocasts of recent humans that were digitized by laser scanning, (i.e., virtual replicas of the physical endocasts that were previously analysed, and four endocasts of fossil specimens reconstructed from CT data). One additional specimen was Cro-Magnon 1, whose virtual endocast was the source of the prototype that was used to obtain the physical measurements. The complete procedure of delimitation of the lateral and upper portions of the endocasts and the quantification of the surface of this area with Avizo 6 on the virtual models was repeated twice. The mean absolute error between the protocol used in this study on the physical endocasts and the quantification of the virtual models was less than one percent. The surface of the frontal, parieto-temporal and occipital lobes are designated below as FALS, PTALs and OALs, respectively.

Different morphometric and statistical procedures were used to analyse the collected data using PAST 2.11 software (Hammer et al., 2001). Linear regressions were calculated using the Reduced Major

Axis algorithm (Miller and Kahn, 1962), which minimizes the errors in both variables (Smith, 2009). All morphometric data were scaled relative to each individual's endocranial volume in order to limit bias in total brain size variation between samples. Thus, the cube-root of the endocranial volume (EV) for each individual was used for size-correction of surface quantification ($(\sqrt[3]{xi})/(\sqrt[3]{EVi}) \times 100$). A principal components analysis (PCA) was used to help visualize the variations on the surface of the endocranial lobes. Percentages of the contribution of the frontal, parieto-temporal and occipital lobes to the constitution of the hemispheres were also calculated. In these analyses, the surface of the hemispheres included those of the frontal, parietal, temporal and occipital lobes on both sides. When a side was not complete enough to quantify the surface of a lobe, the value of the opposite side was doubled to calculate the total surface of the lobes. The coefficient of variation ($CV = SD/mean$) was corrected for small sample size using the parameter V^* , which is calculated as $[(1 + 1/4N) \times CV]$ and expressed in percentages (Sokal and Braumann, 1980; Wood and Lieberman, 2001). The permutation t -test was preferred to the classical t -test to compare the means of the size-corrected surfaces of the various lobes between the different samples. This test uses the t -statistic but is non-parametric. The sequential Bonferroni procedure was used for correction for multiple tests (Rice, 1989; Palmer, 1994).

Fluctuating asymmetry (FA) is a pattern of bilateral variation in a sample where the mean of (R-L) is zero and the variation is normally distributed about that mean (Palmer, 1994). Directional asymmetry (DA) is a pattern of bilateral variation in a sample when a side is statistically larger than the other. DA is detected by statistical tests for departures of the mean (R-L) from zero (Van Valen, 1962). Antisymmetry is a pattern of bilateral variation in a sample when a significant difference exists between sides, but when the larger side varies randomly among individuals (Palmer, 1994). Antisymmetry is detected by statistical tests for departures of frequency distributions of (R-L) from normality in the direction of platykurtosis. The values for kurtosis and skewness were calculated using Excel. Kurtosis was compared to separate critical values for platy- and leptokurtosis (values for Equation. (7), Table 5 in Palmer and Strobeck, 2003). The statistical procedure to test for skewness and tables for critical values are detailed in Sokal and Rohlf (1995). Previous work has shown that "tests for skewness and kurtosis when taken together are probably the most useful way to detect departures from normality for metrical traits" (Palmer and Strobeck, 1992: 67). FA, DA and antisymmetry are the three main patterns of subtle departures from symmetry exhibited by a sample of individuals (Van Valen, 1962; Palmer, 2004).

Different classical terms and indices were used to describe asymmetries between the right and left hemispheres, including the frontal, parieto-temporal and occipital lobes. Signed asymmetry is the difference between the right and left side for each petalia in a given individual (R-L). It retains information about the direction of the asymmetry because positive values indicate a right petalia and negative values indicate a left petalia. Absolute asymmetry is the absolute value of the difference between the right and left side for each petalia in a given individual ($|R-L|$). It also corresponds to FA1 (Palmer and Strobeck, 2003). FA4a estimates the variability of one trait within a given sample, and is calculated using the formula $0.798/\sqrt{\text{var}(R_{xi}-L_{xi})}$. FA1 and FA4a are biased when directional asymmetry or antisymmetry (see below) are present (Palmer, 1994; Graham et al., 1998). The parameter FA11 quantifies the asymmetries for all individuals and for all traits, and is calculated using the formula $\sum(\sum|R_{xi}-L_{xi}|)/N$ (Leary et al., 1985; Palmer, 1994; Balzeau and Gilissen, 2010). This parameter is cumulative and is useful for comparisons of the total size of asymmetry between samples. Here, we quantified FA11 as the sum of the size-corrected values for the surface of the frontal, parieto-temporal and occipital lobes.

Results

Does brain size matter?

We first examined the possible relationships (Fig. 1) between the surface of the frontal, parieto-temporal and occipital lobes (expressed as the square root of the surface of the lobes, and noted as, respectively, SR-FALS, SR-PTALS, and SR-OALS) and endocranial volume (expressed as its cube root, and noted as CR-EV). The correlation was always significant (p always < 0.001) for the frontal, parieto-temporal and occipital lobes (Table 2), using the fossil hominin sample, the AMH sample or the whole combined sample. In details, some variations were visible between the different samples. Indeed, the slope of the significant correlation between the surface of the lobes and endocranial volume was always larger in AMH compared with fossil hominins (Table 2). This trend may be related to the fact that the square-root of the surface of the different lobes increases at a higher rate with increase in cube-root of the endocranial volume. Therefore, the observed difference is due to the more globular shape of the endocranial volume of AMH compared with other fossil hominins. This relationship between the surface of the lobes and the endocranial volumes also explains why **the slope is larger for the parieto-temporal lobes, which are the more extended and rounded area of the upper and lateral portions of the external surface of the brain, for the whole hominin sample.**

Based on these results, particularly the highly significant correlation between the surface of the different lobes and endocranial volume (Table 2), we used relative metric data ($(\sqrt[3]{xi})/(\sqrt[3]{EVi}) \times 100$) to analyse the morphological variation in subsequent analyses to avoid as much as possible variations solely related to changes in the total size of the brain.

Variation in regional brain surfaces during hominin evolution

The distribution of the size-corrected surfaces of the frontal, parieto-temporal and occipital lobes illustrates some variation among the samples (Fig. 2 and Table 3). The distribution observed for the frontal lobes in the Neandertal sample was in the upper end of the range of variation of the other samples, with the highest

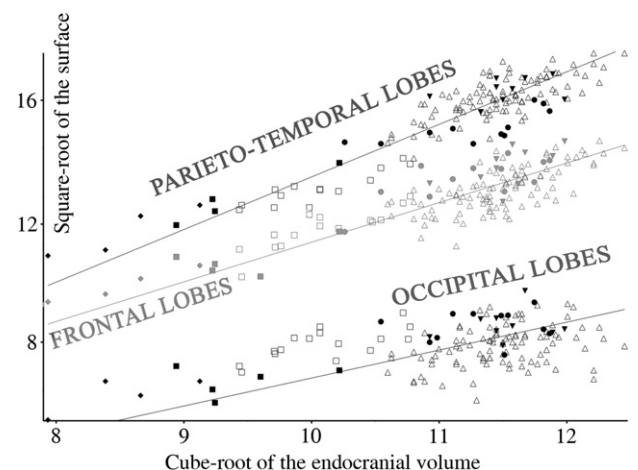


Figure 1. Relationship between the cube-root of the endocranial volume and the square-root of the surface of the frontal, parieto-temporal and occipital lobes in hominins. Black line: linear regression calculated with reduced major axis algorithm. Filled diamonds: *Homo habilis* s.l., filled squares: Georgian and African *Homo erectus*, squares: Asian *H. erectus*, filled circles: Neandertals s.l., inverted filled triangles: fossil anatomically modern *Homo sapiens*, triangles: extant modern humans. Symbols for each individual values are in light gray for the frontal lobes to distinguish them from the values for parieto-temporal lobes that overlap.

Table 2

Regression results for the relationship between endocranial volume and the surface of the frontal, parieto-temporal and occipital lobes.

	Sample	n	Slope	Intercept	R squared	p
Frontal lobes	Complete sample	146	1.310	−1.798	0.739	***
	Fossil hominins	36	1.361	−1.948	0.903	***
	AMH	110	1.928	−9.009	0.452	***
Parieto-temporal lobes	Complete sample	142	1.739	−3.930	0.846	***
	Fossil hominins	33	1.308	0.192	0.910	***
	AMH	109	1.750	−3.970	0.516	***
Occipital lobes	Complete sample	145	0.925	−2.430	0.230	***
	Fossil hominins	36	0.933	−1.754	0.691	***
	AMH	109	1.744	−12.088	0.109	***

***indicates p value <0.001 after sequential Bonferroni procedure for correction for multiple tests.

mean value. The small sample of African and Georgian *Homo erectus* s.l. (sensu lato) has the lowest mean value observed for the size-corrected surface of the frontal lobes and a distribution in the lower end of the range of variation of the other fossil hominin samples. The mean values for this feature in the AMH, Neandertal and Asian *H. erectus* samples are all intermediate, with the AMH sample having the lowest value in this group. While a pattern is present, only the differences in the means between Neandertals and AMH and between Neandertals and African/Georgian *H. erectus* are statistically significant (Table 4).

The lowest values for the surface of the parieto-temporal lobes were found in Asian *H. erectus* (Fig. 2), while the AMH sample has the highest mean and median values among the samples (Table 3). The differences in these means are statistically significant (Table 4). Finally, the occipital lobes showed the largest relative surface in the Asian *H. erectus* sample and the lowest in AMH, with the differences again being statistically significant (Table 4).

The distribution and variation of these lobe surfaces overlap among the various samples. In fact, most of the fossil specimens included in this analysis fall within the range of variation of the large sample of AMH, except notably for some Asian *H. erectus* for the size-corrected surface of the parieto-temporal lobes (Fig. 2).

To further explore the variation between samples expressed in these data, the three first components of the PCA computed with the size-corrected surface of the frontal, parieto-temporal and occipital lobes (Fig. 3) represented, respectively, 60.8%, 23.3% and 15.9% of the total variance. Representation of individuals' scores along the first three components illustrates a strong difference

between the various samples along PC1, whereas all individual specimens are plotted together with no apparent organization related to taxonomic affinities on PC2 and PC3. Along the first component (Fig. 3), the surface of the frontal and occipital lobes loads positively (with loadings of 0.35 and 0.75, respectively) and the surface of the parieto-temporal lobe loads negatively (with a loading of −0.56). The various samples; i.e., comprising specimens of AMH, African and Georgian *H. erectus*, *Homo habilis* s.l., Neandertals s.l. and Asian *H. erectus*, are distributed in this order from the left to the right of the first component (Fig. 3). Some differences in the relative size and pattern of the various lobes are indeed noticeable between the samples (Tables 3 and 4) and explain the distribution of the specimens in the PCA.

Moreover, clear distinctions between the different samples can be made from these data (Fig. 2 and Table 3). In details, the Asian *H. erectus* sample showed a (Table 4) significantly lower dimension for the mean surface of the parieto-temporal lobes (130) compared with all of the other samples (mean values ranging from 134 to 140, Table 3). This difference explains the position of the Asian *H. erectus* sample at the right of the first component of the PCA (Fig. 3). The Chinese specimens from Zhoukoudian were at the extreme right of the distribution of the Asian *H. erectus* sample in this analysis. The Indonesian specimens come from two different chronological samples but the Trinil-Sangiran and Ngandong-Sambungmacan endocrasts plotted together for the analysed features, to the left of the Zhoukoudian fossils. This is due to slightly larger values for the frontal and occipital lobes and smaller values for the parieto-temporal lobe in the Zhoukoudian sample compared with the Indonesian *H. erectus*.

Interestingly, the two samples with the largest endocranial size showed different brain structures, i.e., different variations in regional brain area and contribution of the different lobes (Figs. 2 and 3). Indeed, the Neandertal and the AMH samples were significantly different for the three analysed features (Table 4), with relatively larger frontal (119) and occipital (76) lobes and relatively smaller parieto-temporal (134) lobes (Table 3) for the Neandertals compared with AMH (with respective mean values of 114, 140 and 69). Furthermore, among the analysed samples, the largest mean value for the relative size of the surface of the parieto-temporal lobes was observed in the AMH sample.

Other significant differences (Table 4) were found in the occipital lobes of Asian *H. erectus*, which are significantly larger (79, that was the maximal observed mean value) than those of the African

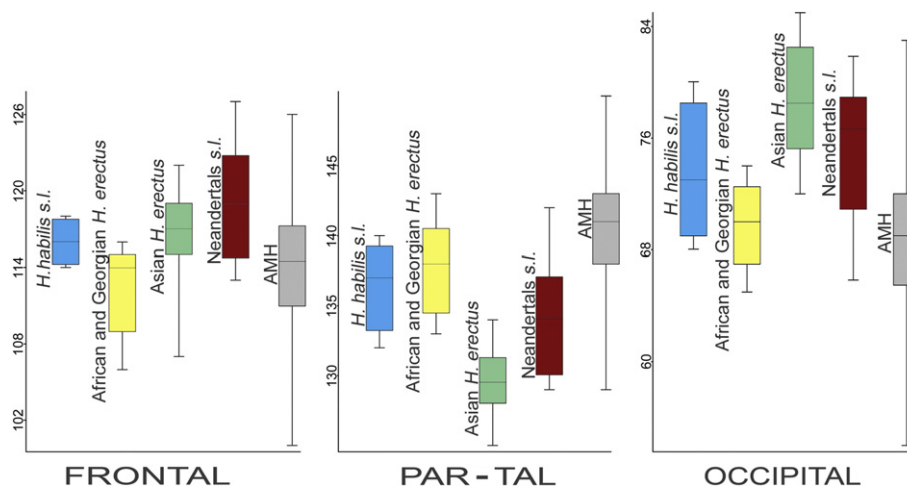


Figure 2. Box plot for the variation of size-corrected surfaces of the frontal, parieto-temporal (PAR-TAL) and occipital lobes in *Homo habilis* s.l., Georgian and African *Homo erectus*, Asian *H. erectus*, Neandertals s.l. and anatomically modern *Homo sapiens*. For each sample, the minimal and maximal values, the 25–75 percent quartiles and the median are illustrated (see also Table 3).

Table 3

Size-corrected dimensions for the surface of the frontal, parieto-temporal and occipital lobes in the different analysed samples.

	<i>Homo habilis</i> s.l.			African and Georgian <i>Homo erectus</i> s.l.			Asian <i>Homo erectus</i>			Neandertals s.l.			AMH		
	n	Mean	V*	n	Mean	V*	n	Mean	V*	n	Mean	V*	n	Mean	V*
Frontal lobes	4	116	1.48	5	113	4.16	15	117	3.16	12	119	4.13	110	114	4.27
Parieto-temporal lobes	4	137	3.32	5	138	3.11	14	130	2.11	11	134	3.50	109	141	2.94
Occipital lobes	4	74	8.20	5	70	5.73	16	79	4.97	12	76	7.27	110	69	8.10

sample of the same species (70) and those of AMH (69). A significant difference was also found between the means of the small frontal lobes of the African and Georgian *H. erectus* (113) compared with Neandertals (119), who have the relatively largest surface of the frontal lobes among the analysed samples.

The variation of the surface of the different lobes may also be characterized as percentages of their contribution to the constitution of the hemispheres, here considered as the sum of the surfaces of the frontal, parieto-temporal and occipital lobes (Table 5). The parieto-temporal lobe was always the major contributor to the total surface of the hemispheres (ranging from 46 to 52% in the analysed samples), followed by the frontal lobes (35–38%) and the occipital lobes (13–17%). Consistent with the previous observations, the contribution of the parieto-temporal lobe was the smallest in the Asian *H. erectus* sample (mean = 46%). This sample also has the largest percent contribution of the occipital lobes (17%) to the total surface compared with the other analysed samples.

Finally, these complementary analyses also illustrate that the parieto-temporal lobe was the one for which the surface showed the least variability within each sample (except when compared with the frontal lobes in the *H. habilis* sample). The sample size-corrected coefficient of variation (V*) for the parieto-temporal lobes in all samples is smaller than those for the samples of the frontal and occipital lobes for the size-corrected surface of the different lobes (Table 3).

It therefore appears that the relative size of the surface of the frontal, parieto-temporal and occipital lobes show relatively limited variation in the genus *Homo* (Fig. 2). However, these features present some complex variation and particularities among the different samples (Tables 3 and 4, Fig. 3) that are not simply related to the overall increase in endocranial volume (Fig. 1) during evolution.

Asymmetries of the brain in AMH

Mean values of the size-corrected classical indices used to describe asymmetries ((R-L), |R-L| and Fa4a) and tests for departure from normality and for directional asymmetry are described in Table 6 for the complete sample of extant anatomically modern *H. sapiens*. The surface of the right hemisphere was significantly larger than the one of the left hemisphere. The distribution of (R-L) this feature was also significantly leptokurtic. Specifically, the right frontal lobe was larger than the left at the population level, but the difference was non-significant. The right parieto-temporal lobe was

significantly larger than the left, and the left occipital lobe was larger than the right. Concerning directional asymmetry, the DA index (which corresponds to mean (R-L) of a given sample) may be compared with another index, FA4a ($0.798\sqrt{\text{var(R-L)}}$). Here, DA was always smaller than FA4a (Table 6). Nevertheless, the parieto-temporal and occipital lobes exhibited significant directional asymmetry. The distribution for the frontal lobes was leptokurtic and the distribution for the parieto-temporal and occipital lobes did not show significant departure from normality.

We also analysed these asymmetries as non-metric traits. The bilateral variations of the surface of the hemispheres and of the frontal lobes were characterized by equal distribution between the right and left sides (respectively, 54%R/46%L for the hemispheres and 48%R/52%L for the frontal lobes). A larger surface for the right parieto-temporal lobe was observed in 63% of the individuals and the proportion of specimens with larger left occipital lobes represented 68% of the samples. These two last observations were associated with significant directional asymmetry (Table 6).

A feature of interest in paleoneurology is the distribution of the frontal and occipital petalias (e.g., LeMay, 1976; Holloway and De La Coste-Lareymondie, 1982; Balzeau et al., 2012). We observed here some interesting patterns. In addition to the results described above, the distribution of the association between the dominant side for the frontal lobes and the dominant side for the occipital lobes was 36% of R/L, 32% of L/L, 20% of L/R and 12% of R/R.

These results on asymmetries in regional brain surface in AMH were not significantly different when the few fossil anatomically modern *H. sapiens* were included in the sample. However, as previously observed (Grimaud-Hervé et al., 2011), the fossils showed larger absolute and relative values for the surface of the different lobes compared with the extant sample. In relation to the larger total size and larger endocranial volume, the fossil sample had larger values for the different indices used to quantify asymmetries but shared with extant humans a similar pattern of asymmetry. Additional work on the evolution of the *H. sapiens* brain is still necessary to better understand these variations in size, shape and gradient of asymmetries.

Asymmetries in regional brain surface: characteristics of fossil hominins

The heterogeneous composition of the fossil hominin sample does not allow a detailed statistical analysis of the variables

Table 4

Statistical test of the means of the size-corrected surface of the frontal (F), parieto-temporal (P) and occipital (O) lobes between the different samples.

	African and Georgian <i>Homo erectus</i> s.l.			Asian <i>Homo erectus</i>			Neandertals s.l.			AMH		
	F	P	O	F	P	O	F	P	O	F	P	O
<i>Homo habilis</i> s.l.	n.s.	n.s.	n.s.	n.s.	**	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.
African and Georgian <i>Homo erectus</i> s.l.				n.s.	**	**	*	n.s.	n.s.	n.s.	n.s.	n.s.
Asian <i>Homo erectus</i>							n.s.	*	n.s.	n.s.	**	**
Neandertals s.l.										**	**	**

n.s. indicates non-significant result, ** indicates p value <0.01 , * <0.05 after sequential Bonferroni procedure for correction for multiple tests.

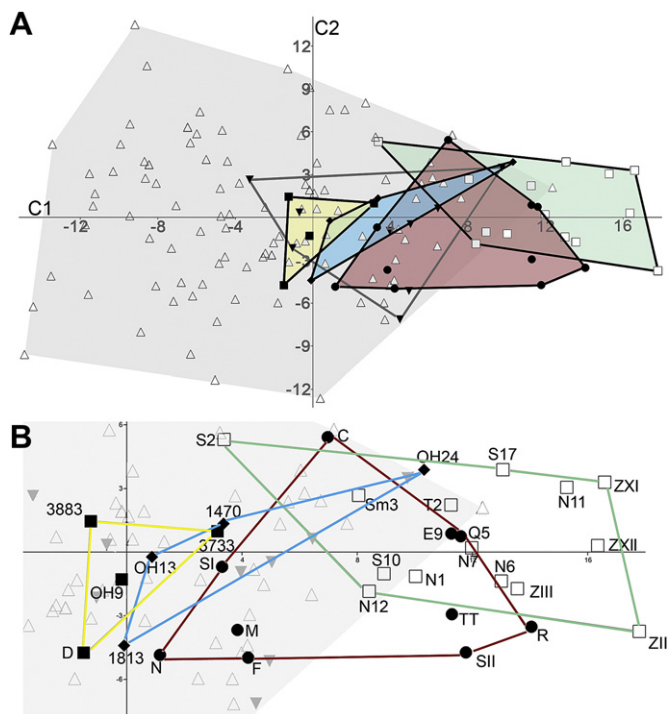


Figure 3. First and second principal components (representing, respectively, 60.8% and 23.3% of the total variance) of the PCA computed on the size-corrected surfaces of the frontal, parieto-temporal and occipital lobes. A: projections of the complete samples. B: focus on fossil hominins with individual labeling. Filled diamonds: *Homo habilis* s.l. (OH13 and 24; 1470, 813; KNM-ER 1470 and 1813), filled squares: Georgian and African *Homo erectus* (D: Dmanisi 2280; 3733, 3383; KNM-ER 3733 and 3883, OH9), squares: Asian *H. erectus* (S2, 10, 17; Sangiran 2, 10 and 17; T2: Trinil 2; N1, 6, 7, 11, 12; Ngandong 1, 2, 6, 7, 11 and 12; Sm3: Sambungmacan 3; ZII, III, XI, XII: Ckn. D1.PA.17, Ckn. E1.PA.16, Ckn. L2.PA.99 and Ckn. L3.PA.100), filled circles: Neandertals s.l., inverted filled triangles: fossil anatomically modern *Homo sapiens*, triangles: extant modern humans (C: La Chapelle-aux-Saints; E9: Ehringsdorf H9; F: La Ferrassie 1; Q5: La Quina H5; M: Le Moustier; N: Neandertal; R: Reilingen; SI, SII: Spy 1, Spy 10; TT: Teshik-Tash).

distribution. Nevertheless, we can compare general trends between the fossil *Homo* (excluding *H. sapiens*) and the AMH sample (including the fossil and recent samples).

The parameter FA11 quantifies the asymmetries for all individuals and for all traits. The mean value for the AMH sample was 8.3, whereas the mean for the fossil *Homo* sample was very close with a value of 8.0. FA11 was 8.8 for African fossils ($N = 8$), 7.8 for Asian specimens ($N = 10$) and 7.2 for the European fossil individuals ($N = 8$). A non-significant negative correlation was observed between absolute values for FA11 and the cube-root of the endocranial volume in extinct species of *Homo* ($N = 26$) or among *H. sapiens* ($N = 105$) whereas the correlation was positive and still non-significant for the whole sample ($N = 131$).

Individual values and variation for each of the analysed species did not show any clear particularity. The range of variation, the observed values as well as the laterality of the asymmetries did not distinguish any of the analysed fossil taxa.

Overall, a larger right frontal lobe was observed in 65% of the fossils, the right parieto-temporal lobe was larger in 58% of the specimens, and the left occipital lobe was larger in 63% of the analysed endocasts. The means of the frontal and parieto-temporal lobes were larger for the right side and the left occipital lobe was larger than the right in the fossil sample. Finally, the distribution of the association between the dominant side for the frontal lobe and dominant side for the occipital lobe was 50% of R/L, 15% of L/L, 5% of L/R and 30% of R/R but the sample comprised only 20 specimens for this feature and was heterogeneous in terms of the representation of the different species. The association R/L for the surfaces of the frontal and occipital lobes appeared to be the most represented, as it was the case for AMH.

This general trend concerning asymmetries in regional brain surface was difficult to interpret as we lack data to analyse these features at the population level. However, based on the different results described above, the pattern for fossil hominins did not appear to differ from the pattern that could be described previously for a much larger sample of extant AMH.

Discussion

Brain shape variations in *Homo*

By documenting variations on the surface of the frontal, parieto-temporal and occipital lobes among different species of the genus *Homo*, this study contributes new information about the structure and the shape of the brain during the last two millions years of hominin evolution.

According to our results, specimens belonging to *H. habilis* s.l. as well as one specimen from Dmanisi, D2280, and African fossils attributed to *H. erectus* s.l. are the only samples that do not show any significant difference with AMH for the relative size of the surface of the frontal, parieto-temporal and occipital lobes. These fossil samples nevertheless exhibit smaller values for the parieto-temporal lobe compared with modern humans. Overall, the different lobes of the upper part of the brain have a similar contribution to the total surface in *H. habilis* s.l. and African and Georgian *H. erectus* compared with anatomically modern humans (Table 5). Asian *H. erectus* and Neandertals share similar characteristics compared with the other groups, i.e., large surface areas for the frontal and occipital lobes and small values for the parieto-temporal lobes. However, if these two groups show some overlap for their individual values, the fossils attributed to Asian *H. erectus* are distinguishable from the other samples in terms of brain structure. They indeed show a significantly smaller surface of the parieto-temporal lobe compared with all of the other samples (Fig. 2, Tables 3 and 4). They also have relatively larger occipital lobes (Table 3). Asian *H. erectus* also have the minimum value for the mean of the contribution of the parieto-temporal lobe to the constitution of the hemispheres and the largest mean value for the occipital lobes compared with the other samples (Table 5). If the Asian *H. erectus* were already recognized for their posteriorly projecting occipital lobe (e.g., Grimaud-Hervé, 1997), this pattern of relative reduction of the parieto-temporal lobe compared with other members of the

Table 5
Percentage contribution of frontal, parieto-temporal and occipital lobes to the hemispheres.

	<i>Homo habilis</i> s.l.			African and Georgian <i>Homo erectus</i> s.l.			Asian <i>Homo erectus</i>			Neandertals s.l.			AMH		
	n	Mean	V*	n	Mean	V*	n	Mean	V*	n	Mean	V*	n	Mean	V*
Frontal lobes	4	36	2.94	4	35	10.61	13	37	5.02	10	38	4.89	108	35	7.13
Parieto-temporal lobes	4	50	4.50	4	52	5.44	13	46	5.15	10	47	4.78	108	52	5.50
Occipital lobes	4	14	17.98	4	13	7.45	13	17	6.50	10	15	10.82	108	13	14.79

Table 6

Indices of asymmetry and tests for directional asymmetry and for departure from normality (kurtosis, skewness).

	Hemispheres	Frontal lobes	Parieto-temporal lobes	Occipital lobes
N	100	100	100	100
(R-L) (or DA)	0.64*	0.33	1.30***	−1.43***
SE	2.82	2.98	3.31	3.71
R-L (or FA1)	2.18	2.27	2.82	3.16
SE	1.89	1.95	2.16	2.40
FA4a	2.24	2.37	2.63	2.95
Skewness	−0.10	0.22	−0.21	0.45
Kurtosis	2.11**	0.96*	0.59	0.48

* indicates a *p* value <0.05, **<0.01, ***<0.001 after sequential Bonferroni procedure for correction for multiple tests.

genus *Homo* was so far unknown. Finally, variation between Neandertals and anatomically modern humans are particularly informative. Indeed, these two species are the most encephalized in the human lineage but they show clear differences in the constitution of their brain. The Neandertal brain is closer to the characteristics observed in Asian *H. erectus* for distribution of regional brain surface whereas the modern human brain recalls the organization observed for older members of the genus *Homo* in Africa. These observations have certainly not to be interpreted in terms of phylogenetic proximity or as indicative of evolutionary relationships, but illustrate a variation in the pattern of brain organization in hominins independent of variations in total size that was never reported in earlier studies of endocranial evolution in hominins.

Our results show that the globular shape of the modern human brain is accompanied by an increase in its absolute and relative size and of the contribution of the surface of the parieto-temporal lobe, whereas Neandertals have the relatively largest surface of the frontal lobe in the genus *Homo*. Enlargement of the parietal lobe in AMH was previously noted (Grimaud-Hervé, 1997; Bruner et al., 2003). This enlargement is also related to the clear individualization of the relief on the surface of the parietal lobes (Grimaud-Hervé, 1997). However, Bruner et al. (2003) also discriminated AMH from ‘archaic’ and Neandertal specimens who share a common endocranial model, including variations related to a unique allometric trend. In our study, AMH show the relatively largest parieto-temporal lobes in terms of metric distribution and mean value (Fig. 2, Table 3). These lobes are also significantly larger in AMH compared with Asian *H. erectus* and Neandertals (Table 4). However, variations in the different analysed samples of the genus *Homo* indicate a much more complex evolution of the human brain than a linear and gradual modification in shape related to an increase in endocranial volume in non-*H. sapiens* species (Holloway, 2007; Sousa and Wood, 2007). The discrepancy with the results of Bruner et al. (2003) might be related to their smaller samples size and lesser variability in terms of analysed species that resulted in an incomplete conception of the complexity of the evolution of the brain in *Homo*. Moreover, Wu et al. (2010) reported a lack of geographical patterning between Chinese, Javanese and African *H. erectus* based on morphometric data on the endocast. Their analyses were based on linear dimensions between intracranial points (endobregma, endolambda) and projecting points of the endocast in defined orientations. Therefore, a noticeable divergence exists between intracranial variation, as observed by Wu et al. (2010), and the morphological variation of endocranial traits directly related to brain anatomy in *H. erectus* s.l. as quantified in our study. Finally, our results challenge the “unexpected stability in anterior brain morphology over the period during which modern human cognitive capacity emerged” (Bookstein et al., 1999: 217). The internal outline of the frontal bone in the mid-sagittal plane may indeed remain stable in the sample of 16 modern humans and

five mid-Pleistocene *Homo*, as studied by Bookstein et al. (1999). However, the features they analysed concern the mid-sagittal curvature of the internal surface of the frontal bone, which is certainly not representative of all possible morphological changes and variations encountered by the anterior brain in the genus *Homo*. Our results here, as well as those of other studies (e.g., Grimaud-Hervé, 1994; Balzeau et al., 2005; Balzeau, 2007; Bruner and Holloway, 2010), demonstrate that the frontal lobes show obvious and discriminating variations during hominin evolution. Undoubtedly, research on endocasts needs to be built upon anatomical landmarks or references that are directly related to brain anatomy instead of points on the internal surface of the bone if one wants to discuss morphological changes or variations of the brain.

Asymmetries of the *H. sapiens* brain

We observed a clear pattern of asymmetry characterizing the surface of the endocranial lobes in AMH. Specifically, the surface of the hemispheres, considered as a whole, exhibits significant directional asymmetry and significant platykurtosis (Table 6). The right hemisphere has a larger surface compared with the left side. The combination of directional asymmetry toward the two sides for the various lobes certainly induces some antisymmetry for both hemispheres. Indeed, significant directional asymmetry is present respectively in favor of the right side for the parieto-temporal lobe and in favor of the left side for the occipital lobe. Few comparative data are available for these features in the literature, which has focused on the asymmetry of linear data or volumetric measurements in pathological perspectives or in relation with manual laterality. Interestingly, asymmetries of the volume of the different lobes measured by means of MRI data in a large sample of modern humans are not in full agreement with our results. Geschwind et al. (2002) found significant directional asymmetry in favor of the right side for the frontal and temporal lobes and in favor of the left side for the parietal and occipital lobes. Similarly, our results pertaining to the surface of the lobes are not significantly asymmetric when considering the frontal lobes and the occipital lobes are significantly asymmetric in favor of the left side. We also observed significant directional asymmetry in favor of the right side for the parieto-temporal lobe that is not directly comparable to the data of Geschwind et al. (2002). Finally, we found a significant asymmetry in favor of the right side for the surface of the hemisphere (corresponding to the surface of the frontal, parieto-temporal and occipital lobes) whereas Geschwind et al. (2002) found a larger non-significant volume for the left hemisphere (that corresponds to the volume of the frontal, parietal, temporal and occipital lobes). Beyond the differences in analysed features, methodologies and samples, the discrepancy between our results and the results of Geschwind et al. (2002) is interesting. The exposed surface of the parietal lobes on the external surface of the brain represents a small proportion of its boundaries. Moreover, the parietal lobes have a relatively flattened external surface, particularly when compared with the frontal or occipital lobes. It has also been shown that the upper and medial part of the left parietal lobe has a significantly more pronounced vertical extension compared with the right lobe in modern humans (Combès et al., 2011; Grimaud-Hervé et al., 2011). This may have some influence on bilateral variations of the volume of the parietal lobes. Similarly, the surface of the temporal lobes certainly does not entirely reflect the variation of the volume of the mid-lateral parts of the brain. Therefore, variations in surface of the lobes are certainly not a good indicator of their variations in total size or volume. Based on the results presented here, the larger left hemisphere, in volume, appears to have a smaller external surface for its upper and lateral portions compared with the right hemisphere.

In this context, data for the surface of the lobes measured on endocasts should not be used to discuss variations in the volume of the lobes of the brain or hemispheric variations. Differences in two- and three-dimensional data for the occipital lobes or for the cerebellum are certainly less biased in relation to the globular shape of these areas. In any case, it seems evident that information acquired on endocasts should not be used to analyse variations in the volume of the different parts of the brain whose intraparenchymal limits are impossible to determine when considering the external surface only.

Asymmetric pattern of the surface of the endocast and right-handedness

After the works of LeMay (e.g., LeMay, 1976, 1977; Galaburda et al., 1978; LeMay and Kido, 1978; LeMay et al., 1982), the pattern of petalias was proposed to be related to manual preferences. However, while LeMay originally considered only the antero-posterior projection of the frontal and occipital lobes, later studies have used the term petalia to refer to lateral lobar extension (e.g., Holloway and De La Coste-Lareymondie, 1982), as well as to measures of bilateral brain width or to volumetric inter-hemispheric variations (e.g., Hopkins and Marino, 2000; Good et al., 2001; Pilcher et al., 2001; Watkins et al., 2001; Hopkins et al., 2008). In living people, a ratio of 8:1 or 9:1 is observed between right and left-handers (e.g., Halpern et al., 2005). However, the correlation between right-handedness and right frontal-left occipital petalias is certainly less clear than previously proposed. Moreover, we recently showed that the petalial pattern of asymmetries (considered in their original sense, i.e., referring to the protrusions of one hemisphere beyond the other) is similar between great apes and hominins, leaving the gradient in the degree of asymmetry as the only relevant structural parameter (Balzeau et al., 2012). When all of the available information about global asymmetries of the brain in modern humans are considered, it appears that the left occipital lobe is significantly larger for its volume (Geschwind et al., 2002), its surface (this study) as well as for the posterior extension of its pole (Holloway and De La Coste-Lareymondie, 1982; Balzeau et al., 2012) than the right occipital lobe. At the same time, the right frontal lobe has a significantly larger volume (Geschwind et al., 2002), a non-significantly larger surface (this study), a non-significantly larger anterior extension of its pole (Balzeau et al., 2012) and a more lateral extension (Holloway and De La Coste-Lareymondie, 1982). Variations in morphology and size of the anterior part of the frontal lobe are related to variations in surface, particularly of the anterior extension, and probably to a lesser degree to variations of the volume of the whole lobe. The results from various studies and for different features illustrates that the occipital lobes exhibit a pattern of significant directional asymmetry in favor of the left side for its posterior extension at the population level in AMH whereas variations of the shape and of the dimensions of the frontal lobes in our species seem to be mostly related to a pattern of fluctuating asymmetry.

In this context, possible correlations between the pattern of asymmetry of large scale morphological traits and the functions are probably more difficult to demonstrate than previously thought. If some anatomical and structural components of the brain are certainly related to preferences in manual laterality, they have not yet been precisely defined and the correspondence to fine details of the surface of the endocast remains to be demonstrated.

Brain asymmetries in the genus Homo

We lack comparative data to discuss asymmetries at the population level for the different species of the genus *Homo*. The

general trend of diminution of the cumulated asymmetries observed in extinct species or among *H. sapiens* in relation to increasing endocranial volume may be related to complex interactions between variations in surface and volume of brain structures. However, our results do show that the general pattern of surface asymmetry of the frontal, parieto-temporal and occipital lobes in fossil species of the genus *Homo* does not seem to be different from the pattern described for AMH.

In what concerns manual laterality in fossil hominins, demonstration of right-handedness from direct archaeological evidence (e.g., Lozano et al., 2008, 2009; Frayer et al., 2010) is certainly more robust than putative interpretation for one fossil specimen of endocranial shape asymmetries because they only make sense at the population level. These studies (Lozano et al., 2008, 2009; Frayer et al., 2010) for example argue for a percentage of dominance of right-handers in a total of 32 Neandertals s.l., similarly to what is observed in modern humans today. Further work integrating such complementary archaeological data and detailed information concerning anatomical asymmetries and their implications is necessary to have a better understanding of behaviors and capacities of fossil hominins.

Conclusions

This study contributes important new information about the evolution of brain structure and morphology in the genus *Homo*. First, our results do show that the general pattern of surface asymmetry for the different regional brain surfaces in fossil species of the genus *Homo* does not seem to be different from the pattern described for a large sample of AMH, i.e., the right hemisphere has a larger surface than the left, as do the right frontal, the right parieto-temporal and the left occipital lobes compared with the contra-lateral side.

It also appears that the contribution of the frontal, parieto-temporal and occipital lobes shows some variation during hominin evolution. Asian *H. erectus* are here discriminated from all other samples of the genus *Homo*, including African and Georgian specimens that are also sometimes included in the taxon *H. erectus*. Indeed, the Asian fossils have a significantly smaller relative size of the parieto-temporal lobes. This observation is particularly relevant in the context of the debate about the inclusion of African and Georgian fossils together with the Asian record into the *H. erectus* species (e.g., Bräuer and Mbua, 1992; Terhune et al., 2007; Baab, 2008; Balzeau et al., 2011). Additional research on the shape and size of the parieto-temporal lobes in *H. erectus* s.l. and new synthetic research on this species will be necessary to further demonstrate how this anatomical area is peculiar among the Asian members of this species and to consider the implication of these possible particularities for taxonomic discussion. Finally, Neandertals and AMH share the largest endocranial volumes of any hominin, but show different brain organization when considering the relative contribution of the frontal, parieto-temporal and occipital lobes. These results illustrate variation in the pattern of brain organization in hominins independent of variation in total size that has not been previously reported. The globularization of the brain and the enlargement of the parietal lobes could be considered as derived features uniquely observed in AMH.

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